Review Exercise 5

1. Solve each of the following equations.

(a) $-6a(-5 - 2a) = 0$

(b) $(4b + 11)(3b - 7) = 0$

(c) $4c^2 = 7c$

(d) $9 + 6d + d^2 = 0$

(e) $49f^2 - 140f + 100 = 0$

(f) $147 - 3h^2 = 0$

2. Solve each of the following equations.

(a) $6k^2 + 11k - 10 = 0$

(b) $3m(8 - 3m) = 16$

(c) $(3n - 1)^2 = 12n + 8$

(d) $(5p + 1)(p + 4) = 2(7p + 5)$

3. **(i)** Solve the equation $2x^2 - 11x - 21 = 0$.

(ii) Hence, solve the equation $2(y + 2)^2 - 11(y + 2) - 21 = 0$.

4. **(i)** If $x = 3$ is a solution of the equation $2x^2 - 5x + k = 0$, find the value of k.

(ii) Hence, find the other solution of the equation.

5. The difference between two numbers is 3. If the square of the smaller number is equal to four times the larger number, find the two numbers.

6. Imran's father is x^2 years old while Imran is x years old. In $4x$ years' time, Imran's father will be twice as old as him.

(i) Form an equation in x and show that it reduces to $x^2 - 6x = 0$.

(ii) Solve the equation $x^2 - 6x = 0$.

(iii) Find the age of Imran's father when Imran was born.

7. The length and the breadth of a rectangle are $(2x + 5)$ cm and $(2x - 1)$ cm respectively. The area of the rectangle is three times the area of a square of sides $(x + 1)$ cm.

(i) Form an equation in x and show that it reduces to $x^2 + 2x - 8 = 0$.

(ii) Solve the equation $x^2 + 2x - 8 = 0$.

(iii) Find the perimeter of the rectangle.

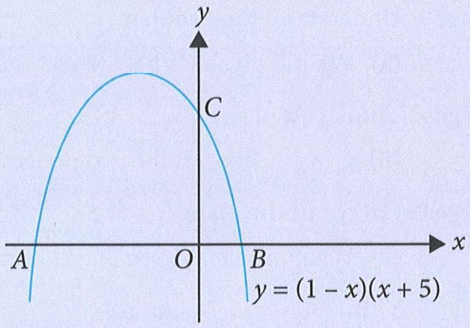
8. If each student in a class sends a New Year greeting card to each of his classmates, a total of 870 cards will be sent. Find the number of students in the class.

9. The height, y metres, of an object projected directly upwards from the ground can be modelled by $y = 20t - 5t^2$, where t is the time in seconds after it leaves the ground.

(i) Find the height of the object 2 seconds after it leaves the ground.

(ii) When will the object be 15 m above the ground?

10. In the figure, the curve $y = (1 - x)(x + 5)$ cuts the x-axis at two points A and B, and the y-axis at the point C. Find the coordinates of A, B and C.



11. The variables x and y are connected by the equation $y = 2x^2 - 9x - 5$. Some values of x and the corresponding values of y are given in the table.

x	−1	0	1	2	3	4	5	6
y	6	−5	p	−15	−14	−9	0	13

(a) Find the value of p.

(b) On a sheet of graph paper, using a scale of 2 cm to represent 1 unit on the x-axis and 1 cm to represent 2 units on the y-axis, draw the graph of $y = 2x^2 - 9x - 5$ for $-1 \leq x \leq 6$.

(c) Use your graph in part **(b)** to find

(i) the coordinates of the turning point,

(ii) the equation of the line of symmetry of the graph.

12. A golfer strikes a golf ball into the air from the ground. The height, h metres, of the ball can be modelled by $h = 32t - 4t^2$, where t is the time in seconds after it leaves the ground.

(i) On a sheet of graph paper, using a scale of 2 cm to represent 1 unit on the t-axis and 1 cm to represent 5 units on the h-axis, draw the graph of $h = 32t - 4t^2$ for $0 \leqslant t \leqslant 8$.

(ii) Use your graph in part **(i)** to find the maximum height of the ball above the ground and the time at which this occurs.

13. A ball is thrown vertically upwards from the top of a school building. Its height, h metres, above the ground, can be modelled by $h = 18 + 6t - 4t^2$, where t is the time in seconds after it leaves the top of the building.

(i) How tall is the school building?

(ii) After how many seconds will the ball strike the ground?



Challenge Yourself

1. Solve the equation $(x - 2)^2 + (x + 3)^2 = 0$, where x is a real number.

Guiding questions (based on **Pólya's Problem Solving Model**)

Stage 1: Understand the Problem

(a) What is given? What are you supposed to find?

Stage 2: Think of a plan

(b) Can you expand the two perfect squares and then simplify and factorise to solve for x?

Stage 3: Carry out the plan

(c) What problem do you face? Can you factorise the quadratic expression?

Stage 2: Think of *another* plan

(d) $(x - 2)^2$ and $(x + 3)^2$ are perfect squares. What values can each perfect square take?

(e) If we add two perfect squares and the sum is 0, what value can each perfect square take?

Stage 3: Carry out the plan

(f) What problem do you face?

(g) Can $x = 2$ and $x = 3$ occur at the same time? If not, what is the solution?

Stage 4: Look Back

(h) Does your solution make sense? Why or why not?

(i) Can we use a graphical method to solve this problem? What graph will you draw? What can we observe from the graph?

2. Find all the integer solutions of the equation $x^2 + y^2 = 10$.

3. Given that the difference between the solutions of the equation $2x^2 - 6x - k = 0$ is 5, find the value of k.

4. **(i)** If a and b are the solutions of the quadratic equation $ax^2 + bx + c = 0$, show that $a + b = -\frac{b}{a}$ and $ab = \frac{c}{a}$.

(ii) Hence, if p and q are the solutions of the equation $2x^2 + 10x + 7 = 0$, where $p > q$, find the value of $p - q$.